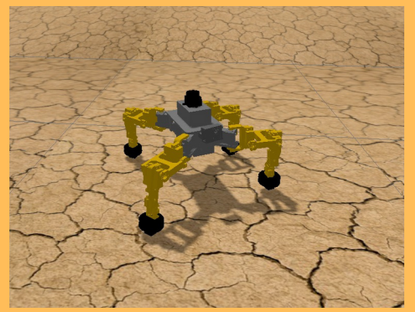
Multimodal robot with bio-inspired gait arbitration across simulated and real environments

Core idea

Hybrid wheel–leg quadruped with three locomotion modes





Validated in Gazebo simulation and on a physical prototype

Method

Multimodal sensing drives neural gait arbitration

Camera LiDAR IMU



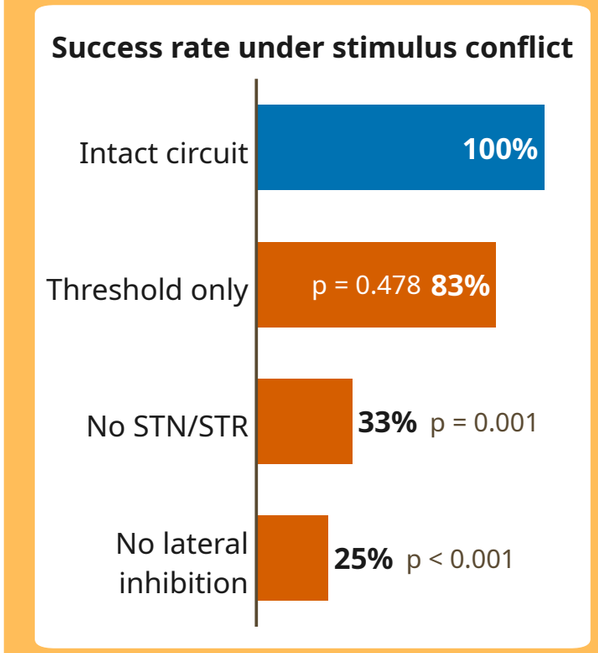
Basal-ganglia winner-take-all circuit

Quadrupedal · Differential · Omnidirectional



Ablation study

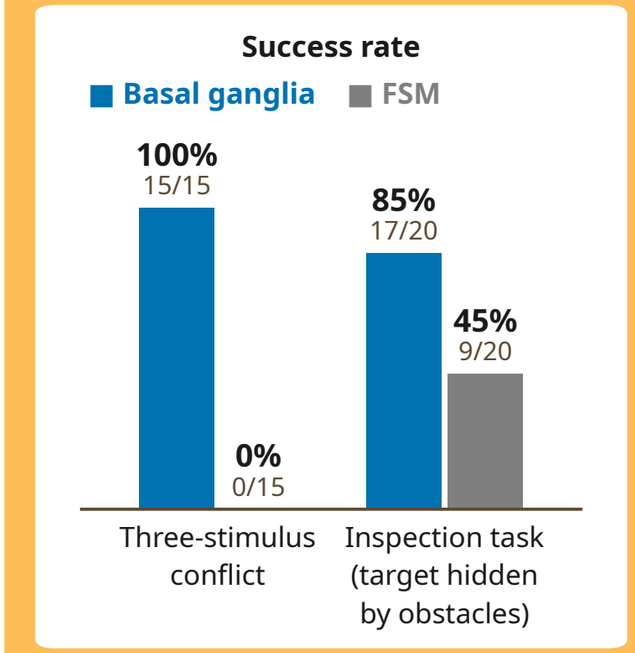
Removing basal-ganglia mechanisms breaks conflict resolution



Simulation · n = 12 trials per variant
Fisher's exact p vs. intact circuit

FSM baseline & inspection

Neural arbitration outperforms a priority-rule FSM baseline



Simulation · n = 15 / 20 trials per controller
matched trial conditions